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Firm's Potential for Co-Creation

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Abstract. Co-creation, an active end-users involvement in product development process, has been recognized as an effective way of minimizing risk of misinterpretation of end-users needs and of achieving product success. Furthermore, high level of co-creation has seemed to create high perceived value to products. With an aim to encourage many more firms to involve their end-users in the product development process, presented in this paper is a quantitative tool developed to help the firms evaluate their readiness and potential for co-creation. Important characteristics of firms for co-creation established in an earlier study have been classified with the help of expert opinion into three different levels: must-have, should-have and nice-to-have. The firms that fulfil all the must-have characteristics are ready for co-creation and their potential is evaluated from the should-have and nice-to-have characteristics. An exploratory case study on three shoe manufacturers was conducted for illustration.

Keywords. Co-creation, product development, potential, customer involvement, co-production

Introduction

Economic growth of most firms mainly relies on their ability to successfully identify end-users needs and quickly transform these needs into successful products. Unfortunately, many products introduced in market do not meet end-users expectations and result in failure. To overcome the product failure, benefits of the active involvement of end-users during NPD commonly referred as co-creation has been long recognized by both practitioners and researchers [1]–[3]. Co-creation is defined as, active end-users involvement during one or all phases of firm initiated NPD projects [2], [4], [5]. Furthermore, it is argued that the high level of co-creation (the extensive involvement of the end-users) will fetch more benefits for the firm during the NPD projects [6]–[8]. The researchers and practitioners have been working on different aspects of co-creation to achieve higher level of co-creation, but surprisingly current literature lacks any tool that can be used to find out the potential for co-creation any firm have [1], [9] or to check even if firm is ready to engage successfully in the process of co-creation [10-12]. Therefore, this study undertakes this problem by developing a quantitative tool to find out the level of potential for co-creation that the firm have.

The remaining sections of the paper are organized as follows. In the next section with the help of literature, steps to attain higher levels of co-creation are briefly summarized. In section 2, the proposed model for the study is discussed. Section 3 is dedicated to explaining the methodology of the study while section 4 focuses on results and discussion. The applicability of the developed tool was demonstrated by an

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exploratory study in section 5. Conclusion and future study directions are discussed in the last section.

1. Steos for acshieveing higher levels of Co-Creation

As aforementioned, the high level of co-creation will offer more benefits to both firm and their end-users. However, in reality firms tend to be at the different levels of co-creation [13] and some firms are not yet ready for co-creation and require preparations internally as well as externally [4], [6], [14], [15]. Therefore, a five-step plan is introduced to help firms in achieving greater levels of co-creation. First and foremost, a firm must assess whether it meets the minimum requirements for co-creation [10] because for every firm there are some predetermined competencies required to engage in co-creation [16]. Once the firm has determined that it is ready to engage in co-creation, the firm should assess the potential that it has for co-creation. In other words, what is the maximum level of co-creation that can be achieved by the firm in current settings[1], [9]. After knowing the potential for co-creation, the firm needs to identify what is the most appropriate method to engage in co-creation [10], [17]. There exist several methods and techniques that can be employed by firms for co-creation and these methods are different from each other not only in means of applications but also results and benefits obtained [6]. In the fourth step, the firm will need to examine its current level of co-creation [13]. This can only be done once the firm has already engaged in co-creation. Knowing this information will allow the firms to assess the amount of efforts that will be required to achieve targeted level of co-creation. The last step is to develop the methods and techniques to increase the level of co-creation to fully enjoy the benefits of co-creation.

This study only covers step one and step two out of the five steps introduced due to the limitation of time and resources. In following section the model proposed in this study is discussed.

2. Proposed model

Successful end-users involvement into the process of NPD is not possible without communication. If a firm is willing to engage in co-creation, it must first build a channel to communicate with its end-users [18]. The channel must be two-way, strong, reliable and active [15]. The firm may choose to use multiple channels if needed. Without these characteristics of the communication channel, the firm cannot freely engage into co-creation process and has to spend more efforts and time to successfully co-create [19]. Along with effective communication, previous literature suggests that there are several other characteristics that are important for co-creation but their levels of importance for co-creation are not the same [15] . Some characteristics are very critical for co-creation, and if they are not available, the firm cannot successfully engage in co-creation. Some characteristics are not that critical but are very important for co-creation. Without them, the firm can still co-create, but it will be very difficult and time consuming for the firm to manage the co-creation operations. Some characteristics are good for co-creation and will ease the process of co-creation. Not having them, however, will not pose any difficulties for the firms during the co-creation project. Therefore, characteristics important for co-creation can be classified into three levels of importance for co-creation, as described below.

- **Must-have:** A firm must have these characteristics, if the firm wants to co-create. Otherwise, the firm cannot co-create.
- **Should-have:** A firm should have these characteristics, if the firm wants to co-create. Otherwise, it will be difficult for the firm to co-create but still the firm will be able to co-create.
- **Nice-to-have:** If a firm has these characteristics, it will be an advantage for the firm but not having them will not result in any difficulty during co-creation.

To utilize different importance level of co-creation characteristics, a two-step model was proposed to check readiness and potential for co-creation for any firm. The first step focuses on basic must-have requirements for co-creation. If the firm does not possess these basic requirements, it is not ready for co-creation. If the firm passes these minimum requirements, it will be evaluated for its co-creation potential. The second step focuses on the characteristics that impact the potential for co-creation, and at this step as per the potential score, the firm will be classified as high, medium or low potential firm for co-creation.

To investigate firm readiness and potential for co-creation, seventeen characteristics identified in an earlier study are utilized. These characteristics are not related to firms' technical or operational capabilities, but they focus on the managerial approaches and mindset of the firms instead. Therefore, it can be argued that they are equally applicable to all industrial sectors and product categories. The characteristics identified are summarized in Table 1.

Table 1. List of characteristics important for co-creation.

Screening system	**Two- way-communication channels**	**Multiple communication channels**
Effective IMS	Staff training in customer relation	Manage customer contribution
Openness to Ideas	Manufacturing personalized items	Mass customization experience
R&D activities	Willingness	Effective information sharing
Flexibility	Communication among end-users	Exploitation
Large market share	High satisfaction level	

3. Methodology

Even though literature mentioned several characteristics of firms that are important for co-creation and has also provided indications that their levels of importance are not the same for co-creation but failed to identify, their degrees of importance. Thus, expert opinion has been explored in this study to determine the importance levels of characteristics obtained in an earlier study through literature. These characteristics were converted into a questionnaire and sent to both researchers and professionals (referred as the experts here onward) working in the field of co-creation and new product development. Respondents were asked to rate the importance of these characteristics individually, according to their experiences into earlier mentioned three categories (i.e., must-have, should-have and nice-to-have). Not-related choice was also provided in the questionnaire to filter out characters that the respondents may believe are not related to co-creation potential but were mentioned in literature.

The selection criterion for researchers was a minimum of two published journal papers related to co-creation fields and for practitioners was a minimum of 10 years working experience in NPD or associated fields. To evaluate the responses, this study utilized weighted average method. The simple weighted average approach is useful in dealing with multiple criteria decision problems. It assumes a set of weights indicating the relative importance of each criterion. The weight score of five, three and two were assigned to must-have, should-have and nice-to-have respectively. Not-related was assigned a weight score of zero as characteristics falling in this category have no impact on co-creation potential. The difference among must-have and should-have weight scores were kept high to reflect the respective difference of importance among two categories. For calculating weighted score for each characteristic, Equation (1) was utilized.

$$W_i = \frac{\sum_{j=1}^{4} w_j x_{ij}}{w_{\max} \sum_{j=1}^{4} x_{ij}} \times 100\% \tag{1}$$

$$W_{cj} = \frac{w_j}{w_{\max}} \times 85\% \tag{2}$$

Where,

W_i = Weighted score for characteristic i

w_j = Weight of category j

x_{ij} = Number of votes of characteristic i received for category j

$w_{\max}$ = Maximum weight among categories

W_{cj} = Cut off weighted score for category j

After a weighted score was calculated, each characteristic was then assigned to one of the four categories. A floor for each category was set at 85% of unanimous votes for that category and a ceiling was limited by the floor of a next higher category. According to Equation (2), the cut off weighted scores were 85%, 51% and 34% for must-have, should-have and nice-to-have respectively. As aforementioned, the characteristics falling under the importance level of must-have will be used to investigate a firm's readiness to engage in co-creation. They are the minimum requirements for any firm to engage in co-creation. Should-have and nice-to-have characteristics will be used to investigate the potential for co-creation with assigned weights of three and two respectively. Different Likert scales can be applied to evaluate these characteristics. An average weighted score obtained from these two categories will be then calculated. Any firm scoring 75% or above of maximum potential score will be classified as a high potential firm for co-creation. Similarly, a firm scoring 50% or more but less than 75% of maximum potential score will be classified as a firm with medium potential for co-creation, and lastly a firm scoring below 50% of total potential score will be classified as firm with low potential for co-creation.

Table 2. Summary of responses from both groups of respondents.

Characteristics	Number of responses received				
	Must have	**Should have**	**Nice-to have**	**Not related**	**Totald replies**
Screening system	17	8	1	1	**27**
Effective IMS	11	12	5	0	**28**
Openness to ideas	21	5	1	0	**27**
R&D activities	17	6	4	0	**27**
Flexibility	17	10	1	0	**28**
Two- way-communication channels	17	10	1	0	**28**
Staff training in customer relation	16	10	2	0	**28**
Manufacturing personalized items	5	5	13	5	**28**
Willingness	24	4	0	0	**28**
Effective information sharing	19	7	2	0	**28**
Manage customer contribution	15	10	3	0	**28**
Mass customization experience	1	6	12	9	**28**
Multiple communication channels	7	14	6	1	**28**
Exploitation	8	17	2	1	**28**
Large market share	4	2	9	13	**28**
High satisfaction level	1	10	9	8	**28**
Communication among end-users	9	12	5	2	**28**

4. Results and discussion

The developed questionnaire was emailed to a total of 112 researchers and 43 professionals whose qualification met the criteria. First and second reminders were sent after seven and fifteen days. A total of nineteen and ten responses were received respectively from researchers and professionals. One response from the researchers group was discarded as it contained same answer for all questions i.e., nice-to-have. After summarizing the responses from practitioners and researchers, T-test was conducted to check if there was any significant difference in opinion of both groups of respondents for each characteristic. The test revealed that out of seventeen characteristics the opinion was different significantly on two characteristics willingness and two-way communication. For the remaining fifteen characteristics, the differences in opinions were not significant at 95% confidence level. The reason for the differences in opinion in two groups for these two characteristics can be credited towards the fact that the practitioners emphasized more on practical requirements while the researchers focused on mindset of the firms. Analyzing these two characteristics separately revealed that researchers have placed willingness in must-have category with 100% weighted score while professionals have placed it in should-have category with 84% weighted score. Even though the opinion of the two groups differed but both groups considered willingness very important for co-creation. According to the definition of co-creation, without the firm's willingness the process of co-creation cannot start [4]. Therefore, willingness of firms to co-create is a must-have characteristics for co-creation and combining the responses of the two groups will result in same i.e. classify willingness into must-have importance level. Similarly, two-way communication was ranked low in must-have importance level by researchers and low in should-have importance level by practitioners. Combining the response from both groups of the respondents will keep two-way communication on the high side of should-have importance level that will approximately represent the opinions of the two groups. Therefore, it was concluded that the responses from the two groups can be combined and analyzed together. The combined responses are summarized in Table 2 and calculations were undertaken using the equation (1) and results were summarized according to the set criteria. The results obtained from the calculation are presented in Table 3.

Table 3. Summary of results from all calculations.

Characteristics	Must have	Should have	Nice to have	Not related	Total replies	Maximum possible score	Total obtained score	Weightage ratio	Importance Level
Willingness	24	4	0	0	**28**	140	132	94.3%	Must have
Openness to idea	21	5	1	0	27	135	122	90.4%	
Effective information sharing	19	7	2	0	28	140	120	85.7%	
Flexibility	17	10	1	0	28	140	117	83.6%	Should have
Two way communication channel	17	10	1	0	28	140	117	83.6%	
Screening system	17	8	1	1	27	135	111	82.2%	
R&D activities	17	6	4	0	27	135	111	82.2%	
Staff training in customer relation	16	10	2	0	28	140	114	81.4%	
Manage customer contribution	15	10	3	0	28	140	111	79.3%	
Effective IMS	11	12	5	0	28	140	101	72.1%	
Exploitation	8	17	2	1	28	140	95	67.9%	
Communication among end-users	9	12	5	2	28	140	91	65.0%	
Multiple communication channels	7	14	6	1	28	140	89	63.6%	Nice to have
Manufacturing personalized items	5	5	13	5	28	140	66	47.1%	
High satisfaction level	1	10	9	8	28	140	47	33.6%	
Mass customization experience	1	6	12	9	28	140	47	33.6%	Not related
Large merket share	4	2	9	13	28	140	44	31.4%	

According to the set criteria three characteristics: willingness, openness to ideas, and effective information sharing were classified into must-have importance level. These results comply with the previous literature and can be justified by the definition of co-creation. Similarly, according to the results mass customization experience and large market share were considered not related to co-creation potential by the experts.

5. Case study

The results of the study were analyzed through an exploratory case study of three shoe manufacturing firms and the potential for co-creation for these firms was evaluated. The purpose of the study was to demonstrate the applicability of the developed tool during practical situations. All three firms selected for the study are global with multiple manufacturing units worldwide and are engaged in designing and production of versatile range of products. First firm, referred as Firm A is well known for its sports shoe wear and global market leader for football shoes. Second firm, referred as Firm B is known for its low-cost, high-quality shoes and famous for formal shoes and is market leader in category of school shoes. Last firm, referred as Firm C is known for its luxurious and comfortable leather shoes. The firm is among top brands for leather shoes in Europe.

The goal of the case study was only to demonstrate the applicability of developed tool therefore the data of three case study firms was gathered from different sources on the internet. To begin the process of checking the potential for the co-creation, it was assumed that all of three selected firms have all the must-have characteristics for co-creation and have fulfilled the minimum requirements needed to engage in co-creation. Also, the scale for the exploratory case study was used 0-1. Zero was assigned to characteristics that firms do not possess and one to the characteristics that firm possess. The 0-1 scale was used due to the limited amount of information available for the selected firms. More precise scale can be employed when accurate information is available. The co-creation characteristics of the selected firms are presented along with their potential scores in Table 4.

Table 4. Summary of potential score calculations for three shoe manufacturers.

Characteristics	Firm A		Firm B		Firm C	
Flexibility	1	3	0	3	1	3
Two- way-communication channels	0	3	0	3	0	3
Staff training in customer relation	1	3	1	3	1	3
Screening system	1	3	0	3	0	3
R&D activities	1	3	1	3	1	3
Manage customer contribution	1	3	0	3	0	3
Effective IMS	1	3	1	3	1	3
Exploitation	1	3	1	3	1	3
Communication among end-users	0	3	0	3	0	3
Multiple communication channels	0	3	0	3	0	3
Manufacturing personalized items	1	2	1	2	1	2
High satisfaction level	1	2	1	2	1	2
Mass customization experience	1	2	0	2	0	2
Total score	**27**		**16**		**19**	

Once the characteristics that the selected firms possess have been identified, the scores were given to the firms and calculations were made according to preset values. After calculating Firm A received a final score of 27 out of 33 that is above 75% and was classified as the firm with high potential for co-creation. Similarly, Firm B scored 16 (below 50%) and was classified as the firm with low potential for co-creation and lastly Firm C scored 19 (more than 50% but less than 75%) and was classified as the firm with medium potential for co-creation. Even though the Firm A managed to land into the high potential for co-creation zone but the firm is still missing two important characteristics of two-way communication, and communication among end-users for co-creation. Therefore the firm needs to pay more attention towards its communication methods and platform to improve its co-creation potential. Similarly, Firm B and Firm C landed in low and medium potential ranges for co-creation respectively as they were missing several important characteristics that are helpful in co-creation.

6. Conclusion

Besides steps for achieving high level of co-creation, a two-step quantitative tool has been introduced in this research to help firms evaluate their readiness and potential for co-creation. According to the experts in the field, fifteen out of seventeen characteristics used in this study were found to have influence on co-creation and were classified into three different importance levels for co-creation: must-have, should-have and nice-to-have. The three must-have characteristics have been set as criteria for checking readiness. Unless they all are fulfilled, the firms are not ready for co-creation. The remaining twelve characteristics have been used to evaluate the potential of the firms that are ready. The higher the score is, the higher the potential the firm have for co-creation. The exploratory case study was conducted to illustrate how the tool can be applied. Due to limited access to the required information, the three firms' potential was calculated based on binary scale. Availability of information will improve the evaluation. It is also foreseen that it is possible that a few characteristics may be swung from one importance level to another level, additional responses from the experts will confirm their importance.

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